AHMET HASDEMİR

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Profile Summary

Economics undergraduate at Hacettepe University specializing in **Quantitative Finance, Applied Econometrics, and Machine Learning**. Bridges macroeconomic theory with stochastic calculus, algorithmic risk architecture, and econometric modeling (OLS, state-space systems, time series). Experienced in developing quantitative derivatives pricing engines, portfolio optimization frameworks, and autonomous data pipelines using Python, SQL, and R.

Education

Hacettepe University — Ankara, Türkiye
Bachelor of Science in Economics (B.S.) — **2024 – 2028 (Expected)**
Minor in Mathematics, Department of Mathematics — **2026 – 2028 (Expected)**

- **Relevant Coursework:** Econometrics I (KDRM, OLS Estimation, Gauss-Markov), Calculus & Real Analysis III, Linear Algebra, Machine Learning for Economists, Financial Markets, History of Economic Thought.
- **Academic Focus:** Econometric time series, mathematical economics, numerical optimization, and computational finance.

Experience

DenizBank — Istanbul, Türkiye
Intern — **Dec 2024 – Jan 2025**

- **Global Insight:** Gained a comprehensive understanding of international banking dynamics and global financial trends through exposure to cross-border operations within the DenizBank and Emirates NBD ecosystem.
- **Strategic Collaboration:** Actively contributed to innovative projects by bridging the gap between local execution and global strategy, enhancing analytical thinking and problem-solving capabilities in a multinational setting.

Technical Skills

Quantitative & Modeling:	Econometric Estimation (OLS, EKF State-Space, Time Series), Modern Portfolio Theory (MPT), Value-at-Risk (Parametric & Historical VaR), Stochastic Calculus, Black-Scholes & Garman-Kohlhagen Models, Monte Carlo Simulations.
Programming Languages:	Python (NumPy, SciPy, Pandas, Statsmodels, Scikit-learn, Matplotlib, Plotly), R, SQL, TypeScript/JavaScript.
Developer Tools & Tech:	Linux (Arch/Hyprland), Git/GitHub, Streamlit, Docker, React Native, Pandoc/LaTeX, REST APIs.
Languages:	Turkish (Native), English (Fluent / Professional Working Proficiency).

Selected Quantitative Projects

QuantSuite — Multi-Asset Derivatives & Volatility Analytics — *Python, SciPy, Plotly, Streamlit*
Derivatives Pricing & Volatility Engine — github.com/ahasdemir/QuantSuite

- Architected a multi-asset quantitative platform implementing dividend-adjusted **Black-Scholes-Merton** for equities and the **Garman-Kohlhagen** dual-interest model for FX pairs.
- Built high-speed vectorized computation for first and second-order **Greeks** ($\Delta, \Gamma, \Theta, \mathcal{V}, \rho$).
- Implemented a custom **Newton-Raphson** numerical solver to extract Implied Volatility (IV) and dynamic **3D Volatility Surfaces** (Maturity × Strike × IV) for market skew and tail-risk detection.

Algo-Risk Monitor — Financial Risk & Portfolio Optimization Engine — *Python, Pandas, MPT, Seaborn*
Quantitative Risk Analysis & Stress Testing — github.com/ahasdemir/Algo-Risk-Monitor

- Developed a systematic portfolio risk dashboard pulling automated OHLCV data across S&P 500 equities and major digital assets.
- Implemented **Modern Portfolio Theory (MPT)** with 10,000+ path Monte Carlo simulations searching for the Efficient Frontier and Maximum Sharpe Ratio portfolios.
- Structured a Hull-style parametric and historical **Value at Risk (VaR)** engine and forward scenario projections via **Geometric Brownian Motion (GBM)**.

Dynamic Multi-Compartment State-Space Physiological Model — *Python, OLS, Time Series, EKF*
Applied Econometrics & Parameter Filtering — *Quantitative Research*

- Formulated a 4-compartment dynamic state-space discrete update model ($W_t = \text{FM}_t + \text{MM}_t + \text{Water}_t + \varepsilon_t$) based on Hall and Forbes energy balance dynamics, running multiple linear regressions (OLS) on 70+ days of real-world micro-data.

FitHash — Autonomous Mobile Health & Progression Tracker — *React Native, TypeScript, Android*
Offline-First Mobile Tracking Engine — github.com/ahasdemir/fithash

- Developed a full-stack offline-first mobile application featuring Epley-based 1RM autoregulated progression algorithms, wear sync, and native backup/restore mechanics.

Certifications & Activities

- **Python Certification** — Kaggle (Advanced data manipulation, vectorization, algorithmic scripting).
- **Open Source & Technical Writing** — Authoring quantitative research and technical analysis on personal research website (hasdemir.vercel.app).